

MSAI 699 Capstone Project

Capstone Project Proposal:

**NeuroGuide AI: An Explainable Multi-Agent Artificial Intelligence
Platform for Personalized Cognitive Health Education Using
Machine Learning and Retrieval-Augmented Generation**

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Abstract

Innovation in AI technologies such as natural language processing, machine learning, and large language models have revolutionized access to information about the healthcare field (Bender et al., 2021; Lewis et al., 2020). Nevertheless, general-purpose AI systems produce misinformation because of the absence of mechanisms of validation of their answers. It is particularly problematic when it comes to cognitive health education, as patients and caregivers need factual information about Alzheimer's disease, mild cognitive impairment, genetics, and life style interventions (Alzheimer's Association, 2024; World Health Organization, 2023).

NeuroGuide AI is the proposed solution – an explainable multi-agent artificial intelligence platform that utilizes Retrieval-Augmented Generation (RAG), machine learning, explainable artificial intelligence (XAI), and dedicated AI agents for delivery of personalized cognitive health education. This system will provide retrieval and validation of information about the medical sources, as well as risk assessments for educational purposes with help of machine learning and SHAP feature level explanations (Breiman, 2001; Lundberg & Lee, 2017).

Evaluating this project will be done through metrics of retrieval quality, response grounding, machine learning performance, explainability, and usability of the technology. The purpose of this research is to show that machine learning, RAG, XAI, and multi-agent architecture of the system can increase the accuracy and transparency of AI assisted cognitive health education.

Proposal

Project Title

NeuroGuide AI: An Explainable Multi-Agent Artificial
Intelligence Platform for Personalized Cognitive Health Education
Using Machine Learning and Retrieval-Augmented Generation

Statement of the Project

Large language models have changed the way people access healthcare information by providing The emergence of large language models has brought about a revolution in how people access information regarding their healthcare through conversational answers. In spite of these technological advancements, a lot of AI technologies are still vulnerable to hallucinations, out-of-date information, and lack of transparency (Bender et al., 2021). These problems are especially critical in healthcare because wrong information can adversely affect user understanding and decision-making.

This project is aimed at the development of NeuroGuide AI, an artificial intelligence platform that will consist of several agents and will be capable of providing users with evidence-based cognitive health education. The difference between NeuroGuide AI and other chatbots is that this technology uses Retrieval-Augmented Generation, machine learning, explainable AI, and several AI agents (Lewis et al., 2020).

Educational risk assessment based on machine learning will take into account demographics, lifestyle, cardiovascular, and genetic factors such as APOE ϵ 4 (Livingston et al., 2024).

Explainability using SHAP will give transparent explanations at the level of features for each prediction (Lundberg & Lee, 2017).

This project is based on a combination of several areas of artificial intelligence covered during the Master of Science in Artificial Intelligence curriculum, including machine learning, natural language processing, Retrieval-Augmented Generation, explainable AI, human-computer interaction, software engineering, cloud computing, and responsible AI. After completion, the developed system will be evaluated for its potential to increase the accuracy, transparency, and educational benefits of the AI-assisted cognitive health education as compared to traditional one-agent conversational AI systems.

Research Questions

Primary Research Question

Can an explainable multi-agent Retrieval-Augmented Generation (RAG) architecture provide more accurate, transparent, and trustworthy cognitive health education than a traditional single-agent conversational AI system?

Secondary Research Questions

RQ1: Does a multi-agent architecture improve the quality and reliability of educational responses compared to a single-agent approach?

RQ2: Can machine learning–based educational risk assessments improve personalization while maintaining transparency through explainable AI?

RQ3: How does Retrieval-Augmented Generation improve factual grounding and reduce unsupported responses?

RQ4: How do users perceive the usability, trustworthiness, and educational value of a multi-agent cognitive health platform?

Hypothesis

It is hypothesized that NeuroGuide AI will produce more accurate, transparent, and trustworthy educational responses than a traditional single-agent conversational AI system. By combining Retrieval-Augmented Generation, machine learning, explainable AI, and a multi-agent architecture, the platform is expected to improve factual grounding, reduce unsupported responses, and increase user confidence while providing personalized educational guidance.

Proposed AI Technologies

NeuroGuide AI integrates multiple artificial intelligence concepts explored throughout the Master of Science in Artificial Intelligence program.

Machine Learning

- Random Forest classification (Breiman, 2001)
- Educational risk prediction
- SHAP explainability

Natural Language Processing

- Large Language Models (GPT-4o-mini)
- TF-IDF document retrieval
- Latent Semantic Analysis (LSA)
- Prompt engineering

Retrieval-Augmented Generation

- Hybrid sparse and semantic document retrieval (Lewis et al., 2020)
- Citation-grounded response generation
- Confidence scoring

Multi-Agent Artificial Intelligence

The platform consists of four specialized AI agents (Wu et al., 2023).that independently perform retrieval, reasoning, validation, and response synthesis.

Human-Computer Interaction

- Personalized user interface

- Patient and clinician response modes
- Accessibility-focused interface design

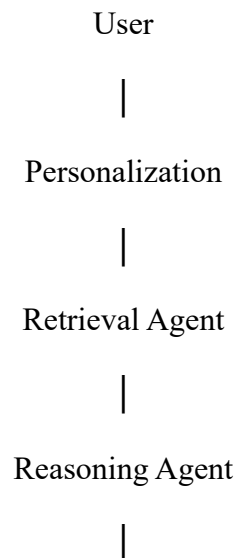
Responsible AI

- Explainable AI (SHAP)
- Bias awareness
- Source transparency
- Educational medical disclaimers

Proposed System Architecture

The NeuroGuide artificial intelligence system analyzes all user requests via personalization step and a set of four different AI agents working together in harmony. In contrast to one language model being used, each particular agent is responsible for a specific task contributing to the high accuracy, integrity, and reliability of the final output.

Model architecture:



Validation Agent

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Synthesis Agent

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User

The proposed architecture consists of four collaborating AI agents.

Personalization

User profile information, including demographic and educational risk factors, is used to personalize educational responses and risk assessments.

Retrieval Agent

Searches trusted cognitive health resources and retrieves the most relevant supporting evidence.

Reasoning Agent

Analyzes retrieved information and generates evidence-based educational explanations tailored to the user's request.

Validation Agent

Verifies that generated responses are supported by retrieved evidence and assigns a confidence score to improve transparency.

Synthesis Agent

Produces the final educational response while adapting language for either patient or clinician audiences and providing supporting citations.

Datasets

The knowledge base will consist of trusted public sources related to Alzheimer's disease and cognitive health, including:

- PubMed Central
- NIH MedlinePlus (National Institute on Aging, 2024)
- Alzheimer's Association (Alzheimer's Association, 2024)
- Cochrane Reviews
- World Health Organization (WHO) (World Health Organization, 2023)
- Centers for Disease Control and Prevention (CDC)

The educational risk assessment model is based on publicly available dementia risk factors summarized by the **2024 Lancet Commission** (Livingston et al., 2024), incorporating demographic, lifestyle, cardiovascular, and APOE $\epsilon 4$ genetic information.

Evaluation

The effectiveness of NeuroGuide AI will be evaluated using quantitative and qualitative methods to assess system performance, machine learning accuracy, explainability, and user experience.

AI Performance

The Retrieval-Augmented Generation system will be evaluated using:

- Retrieval relevance
- Citation grounding
- Response confidence
- Response latency

Machine Learning

The educational risk assessment model will be evaluated using standard machine learning metrics, including:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Explainable AI

Model transparency will be assessed using SHAP feature attribution to verify that educational risk predictions are explainable and clinically reasonable (Lundberg & Lee, 2017).

Human-Computer Interaction

User experience will be evaluated through:

- System Usability Scale (SUS)
- User satisfaction
- Accessibility review
- Response readability

These evaluation methods will determine whether the proposed multi-agent architecture improves the reliability, transparency, and educational value of AI-assisted cognitive health education.

Ethical Considerations

NeuroGuide AI works within the field of healthcare education, which means that ethical AI use is crucial in this project (Bender et al., 2021).

The system will operate only as an educational tool and won't give any diagnoses or suggestions on treatments. All answers will contain disclaimers urging people to seek assistance from licensed healthcare providers regarding any medical concerns.

For additional transparency, all answers will contain citations of reliable medical sources wherever possible. Explanatory approaches such as SHAP will help users comprehend how particular risk factors influence educational risk assessments.

Protected health information will not be collected and stored in the project. Possible sources of biases will be accounted for during the development of the project through using evidence-based medical resources and assessing the fairness of the answers.

Demonstration of Program Learning Outcomes

Program Learning Outcome 1

Apply artificial intelligence principles to solve complex real-world problems.

This project applies machine learning, Retrieval-Augmented Generation, natural language processing, and multi-agent artificial intelligence to address the challenge of providing accurate and trustworthy cognitive health education.

Program Learning Outcome 2

Demonstrate independent research skills using artificial intelligence literature.

The project integrates peer-reviewed research, trusted healthcare resources, and current artificial intelligence methodologies to support system development and evaluation.

Program Learning Outcome 3

Develop and evaluate artificial intelligence solutions.

NeuroGuide AI will be designed, implemented, and evaluated using established performance metrics, usability testing, and explainability techniques to assess its effectiveness as an educational platform.

Program Learning Outcome 4

Evaluate ethical, legal, and societal implications of artificial intelligence.

The project incorporates responsible AI principles through transparency, explainability, source attribution, educational disclaimers, and consideration of bias within healthcare applications.

Program Learning Outcome 5

Communicate artificial intelligence concepts effectively.

The project will include technical documentation, system architecture, evaluation results, and a formal research paper that communicate complex artificial intelligence concepts to both technical and non-technical audiences.

Conclusion

The NeuroGuide AI system is an example of how a number of artificial intelligence approaches can be used together on one educational platform to generate reliable information related to cognitive health care. The use of machine learning, explainable AI, Retrieval-Augmented Generation, and a multi-agent system aims to increase the efficiency and personalization of AI-based health care education in addition to demonstrating knowledge gained during the MSAI program. (Breiman, 2001; Lewis et al., 2020; Lundberg & Lee, 2017).

AI Disclosure: Artificial intelligence was used only to assist with proofreading, grammar refinement, formatting, and organization. The project idea, research direction, system design,

implementation, technical decisions, analysis, and all final content were developed independently by the author, who assumes full responsibility for this submission.

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